

Data Management
2023/2024

Data Warehouse

techniques and tools



SAPIENZA
UNIVERSITÀ DI ROMA

Tutor
Roberto Maria Delfino
delfino@diag.uniroma1.it

Contents

- Motivations
- Data Warehouse recap
 - OLTP vs OLAP
 - The DW stack and Data Marts
 - The multidimensional model
 - Characteristics of the {R,M,H}OLAP models
 - The ETL (Extract, Transform, Load) layer
 - Hierarchies in the DFM model
 - OLAP operations (roll-up and drill-down, slice and dice, pivoting, drill-across)
- Data Warehousing Tools
- Demo

Contents

- Motivations
- Data Warehouse recap
 - OLTP vs OLAP
 - The DW stack and Data Marts
 - The multidimensional model
 - The ETL (Extract, Transform, Load) layer
 - Hierarchies in the DFM model
 - OLAP operations (roll-up and drill-down, slice and dice, pivoting, drill-across)
- Data Warehousing Tools
- Demo



On the role of Data today (and tomorrow)

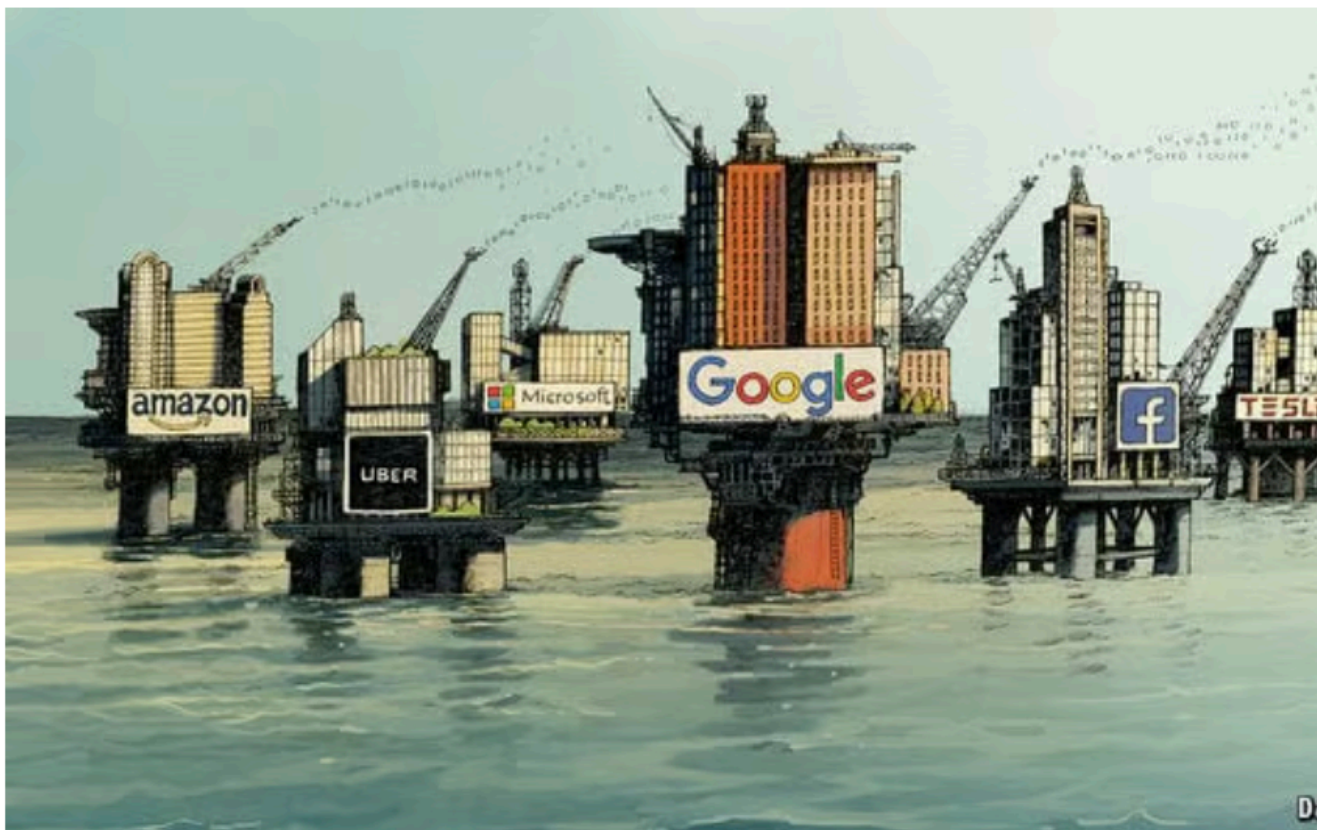


Menu Weekly edition The world in brief Search

Leaders | Regulating the internet giants

The world’s most valuable resource is no longer oil, but data

The data economy demands a new approach to antitrust rules



May 6th 2017

MIT
Technology
Review

Maximize business value with data-driven strategies

Every organization is now collecting data, but few are truly data driven. Here are five ways data can transform your business.

By Matt Hobbs & Anil Nagaraj

June 22, 2022

Forbes

FORBES > INNOVATION > ENTERPRISE & CLOUD

A New Survey Finds Increasing Business Impact Of Data And AI Executives

Data → Information → Knowledge

Data → Information → Knowledge

Data: Collection of symbols with no explicit semantics

Information = Data + Semantics

Knowledge = Information + Application

"Scientia potentia est"
(T. Hobbes)

Whatever the organisation, knowledge is crucial!

Data → Information → Knowledge

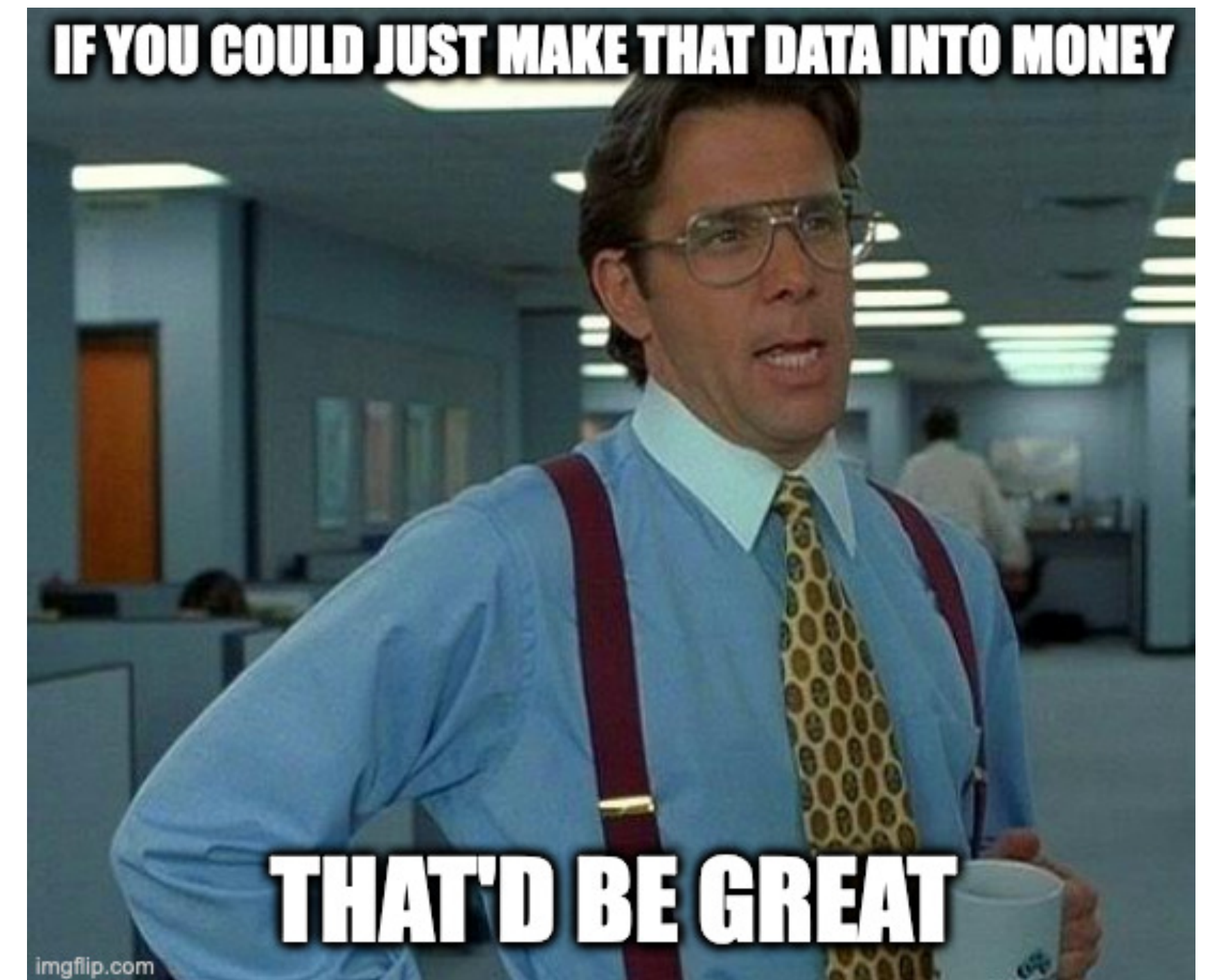
Data: Collection of symbols with no explicit semantics

Information = Data + Semantics

Knowledge = Information + Application

"Scientia potentia est"
(T. Hobbes)

Whatever the organisation, knowledge is crucial!



Data Warehouse

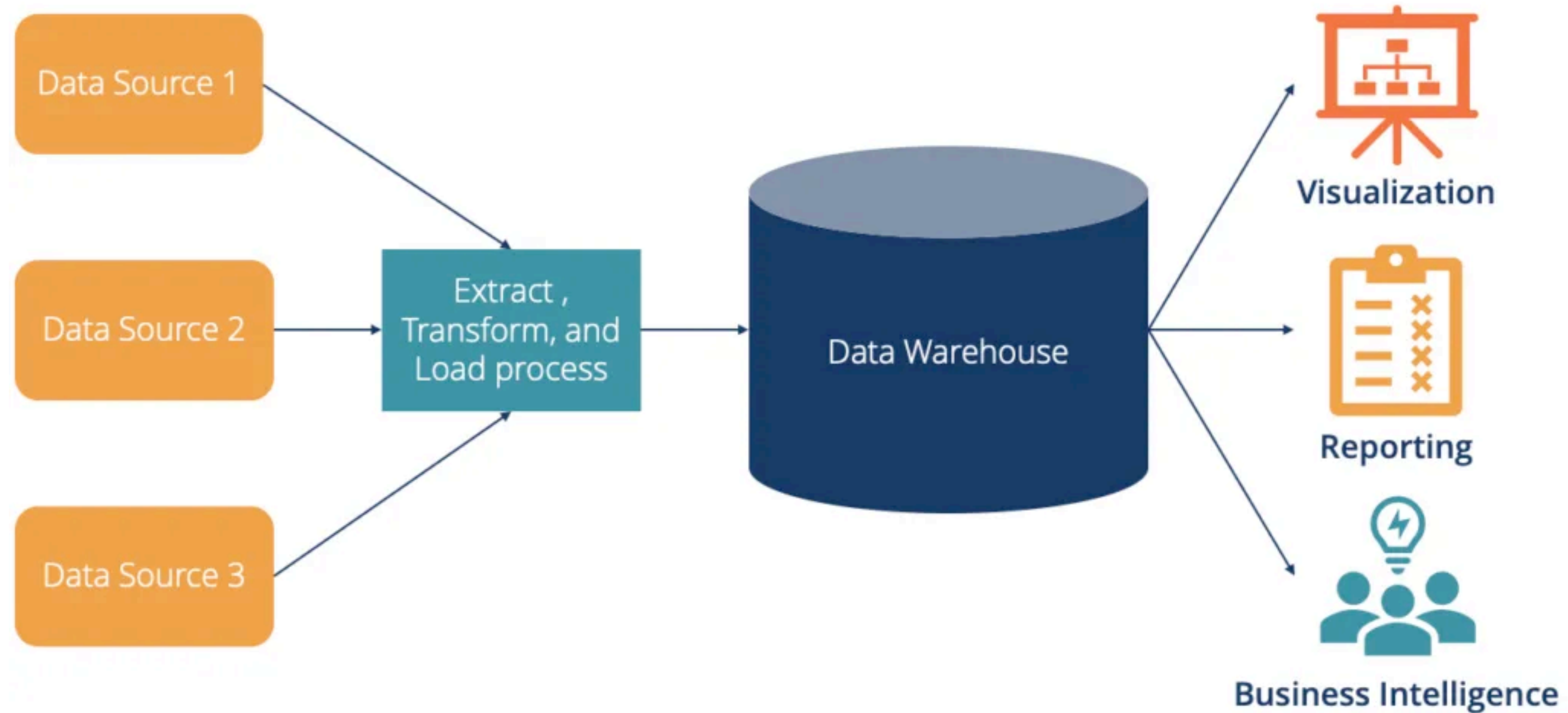
Repository of data to support knowledge workers in strategic decision-making processes

Main pillars:

- Subject-oriented
- Integrated and consistent
- Evolution over time
- Non volatile



The Data Warehouse big picture

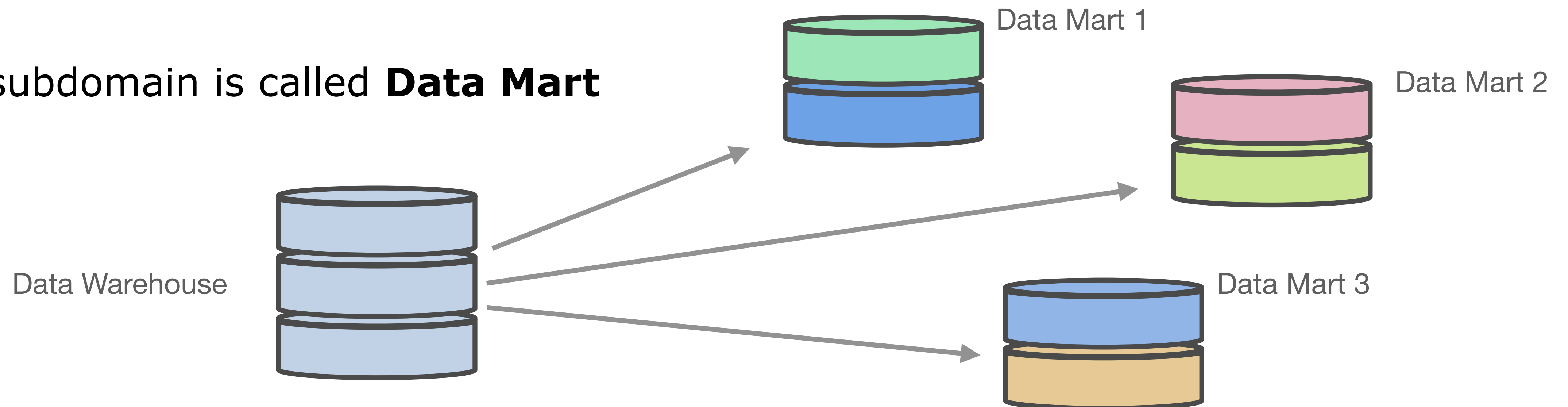


Data Mart

A Data Warehouse generally consists of huge amount of data involving several aspects of a given business domain

In many cases analyses are based on just subsets of the whole domain represented by the data stored in a DW

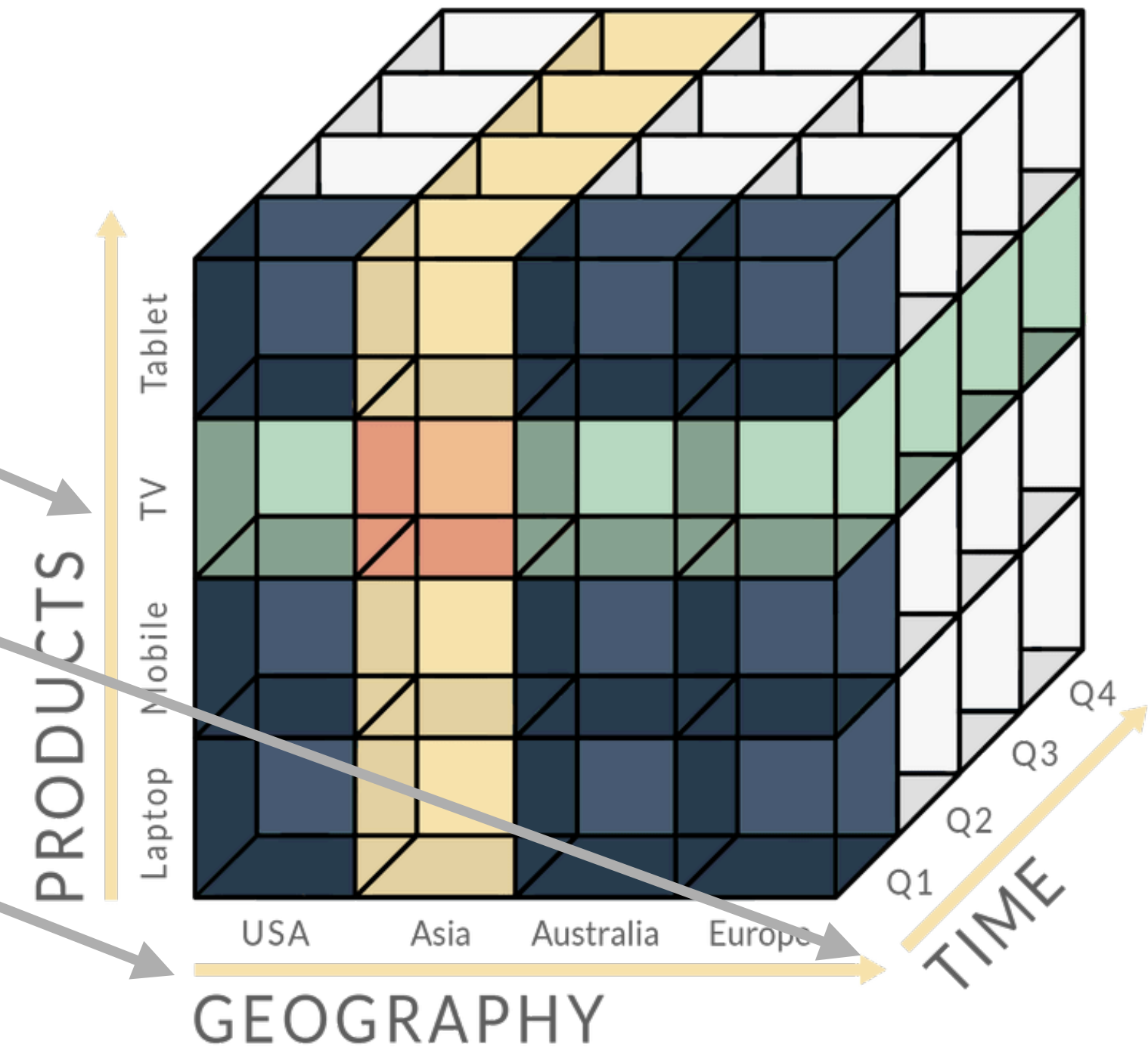
Each subdomain is called **Data Mart**



Multidimensional model

Data virtually represented as hypercubes (dimensions, facts, measures)

ID	Geo	Price	Time	Qty	Prod	Disc
A10	USA	400	Q1	3	Mobile	20%
B13	Asia	600	Q3	7	Tablet	0%
D31	USA	900	Q2	2	TV	18%
C7	Europe	500	Q1	13	Mobile	0%
F9	Europe	500	Q4	5	TV	10%
E11	Asia	1000	Q4	4	Laptop	12%



Multidimensional model - Aggregation

Suppose we want to know the average base price and average discount applied to sold products for continents and quarters

Geo	Time	Prod	avg(Price)	avg(Disc)
USA	Q1	Mobile	841\$	20%
USA	Q1	Tablet	765\$	23%
...	...	...	...	...
Europe	Q3	Mobile	698\$	13%
Europe	Q4	TV	453\$	16%
...	...	...	...	...

```
SELECT Geo, Time, Prod, avg(Price), avg(Disc)
FROM T
GROUP BY Geo, Time, Prod
```

- Dimensions
- Measures

Hierarchies

Axes of hypercubes are called the dimensions and they are used as the coordinates needed to identify portions of the multidimensional data structure containing measures

In some cases dimensions can be organised in **hierarchies**, whose layers allow to express higher or lower levels of granularity in terms of aggregation

Location hierarchy: *Country* → *Region* → *Province* → *City* → *District*

OLAP operations can make use of hierarchies to make analyses based on different levels of detail

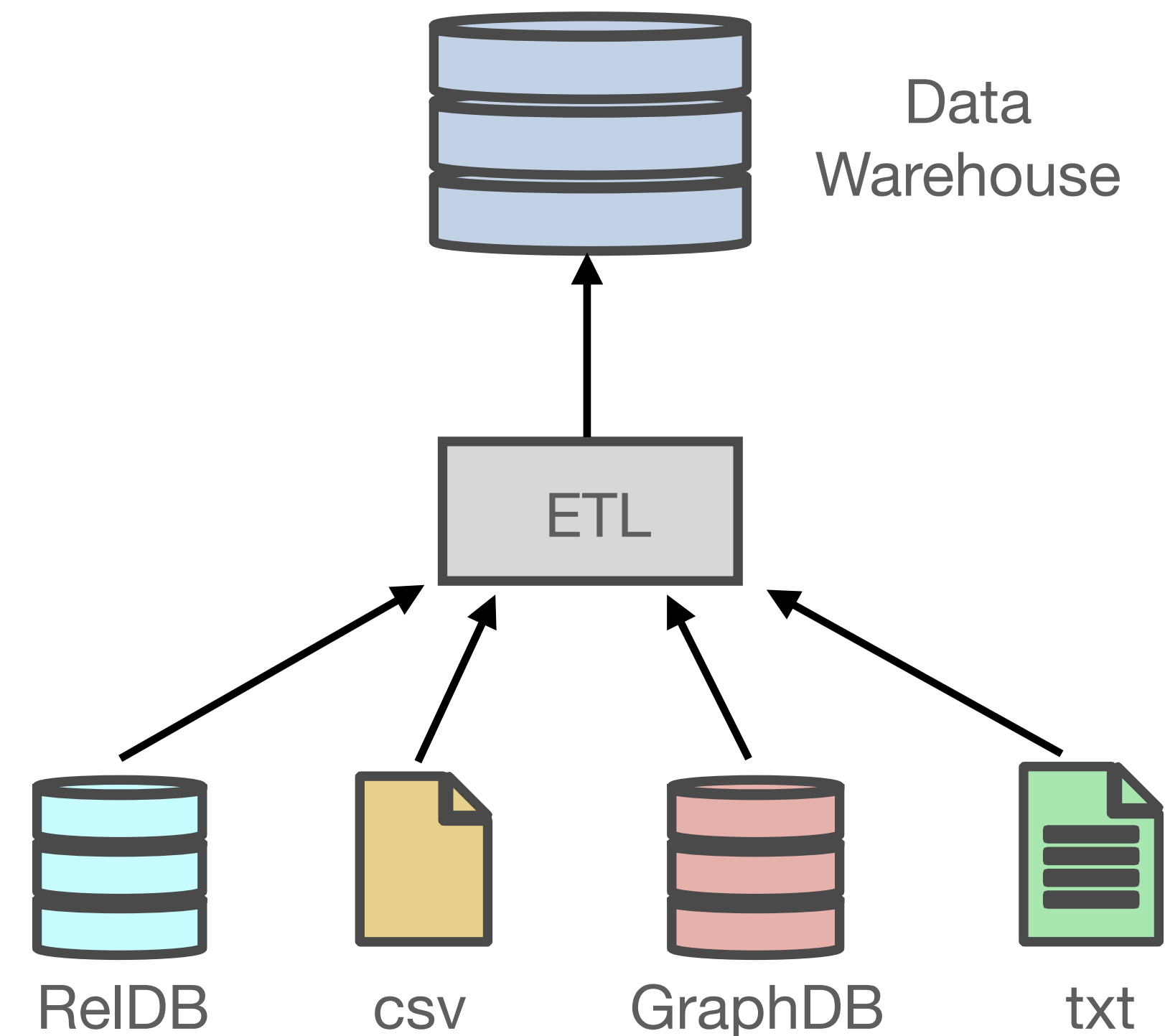
Integration and ETL (Extract, Transform, Load)

Data sources used to populate a Data Warehouse may be highly heterogeneous (relational DBs, csv, XML, unstructured data, etc.)

Integrating such data is not a trivial task:

- missing data
- duplicate values
- incorrect values
- different formats
- entity resolution
- target schema definition

ETL: plethora of techniques to address such problems



OLAP session

Process made of subsequent analytical steps where each step is the result of the outcome of preceding ones

An OLAP session is like a navigation path consisting of facts' analyses based on different view points, obtained by means of sequences of queries which are executed on the basis of the OLAP operators used at each step (e.g., by changing aggregation granularity)

Session example:

S1: average interest rate of mortgages granted in Italy

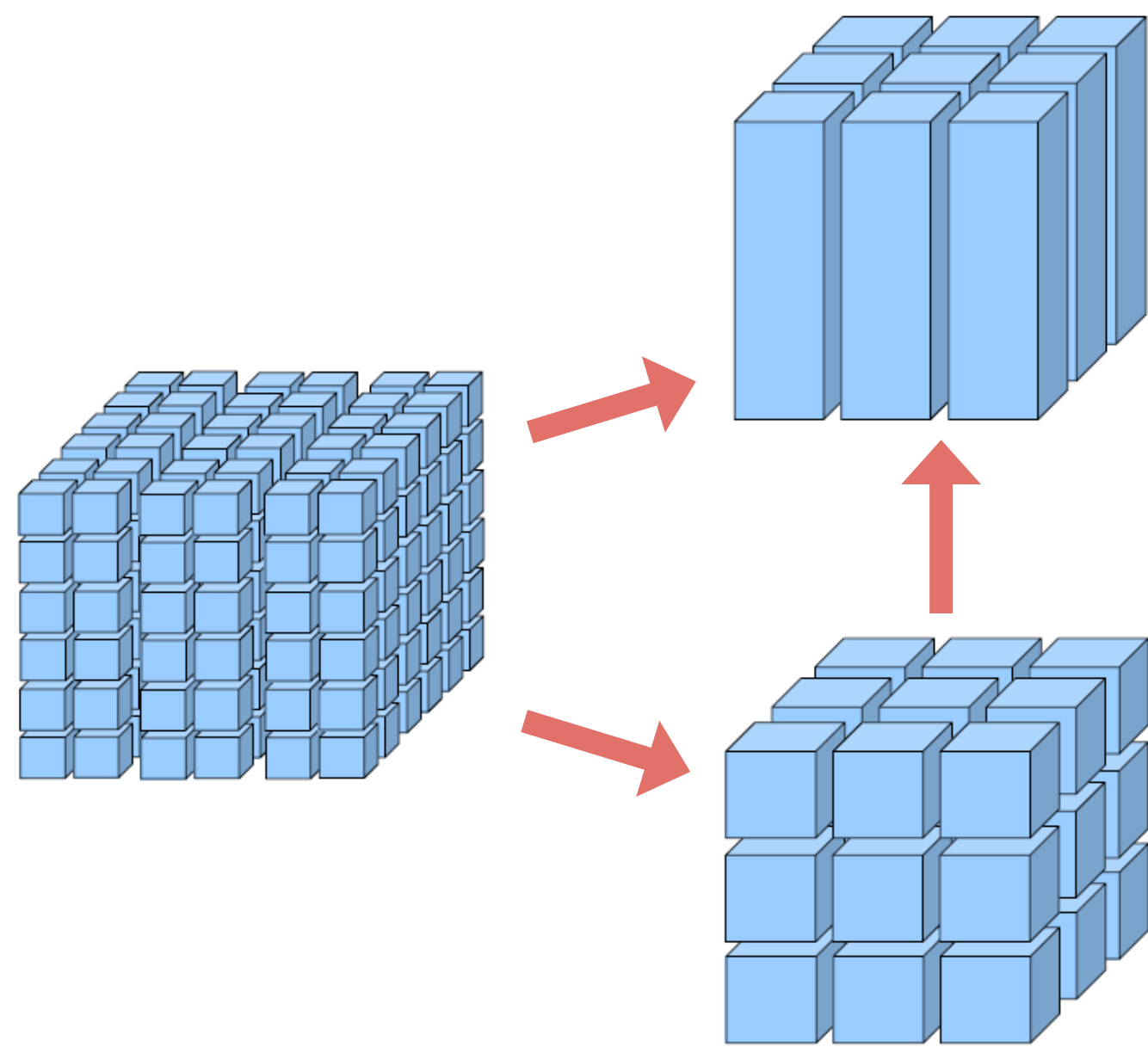
S2: average interest rate of mortgages granted in Italy for each Region

S3: average interest rate of mortgages granted for each Region, for Education level

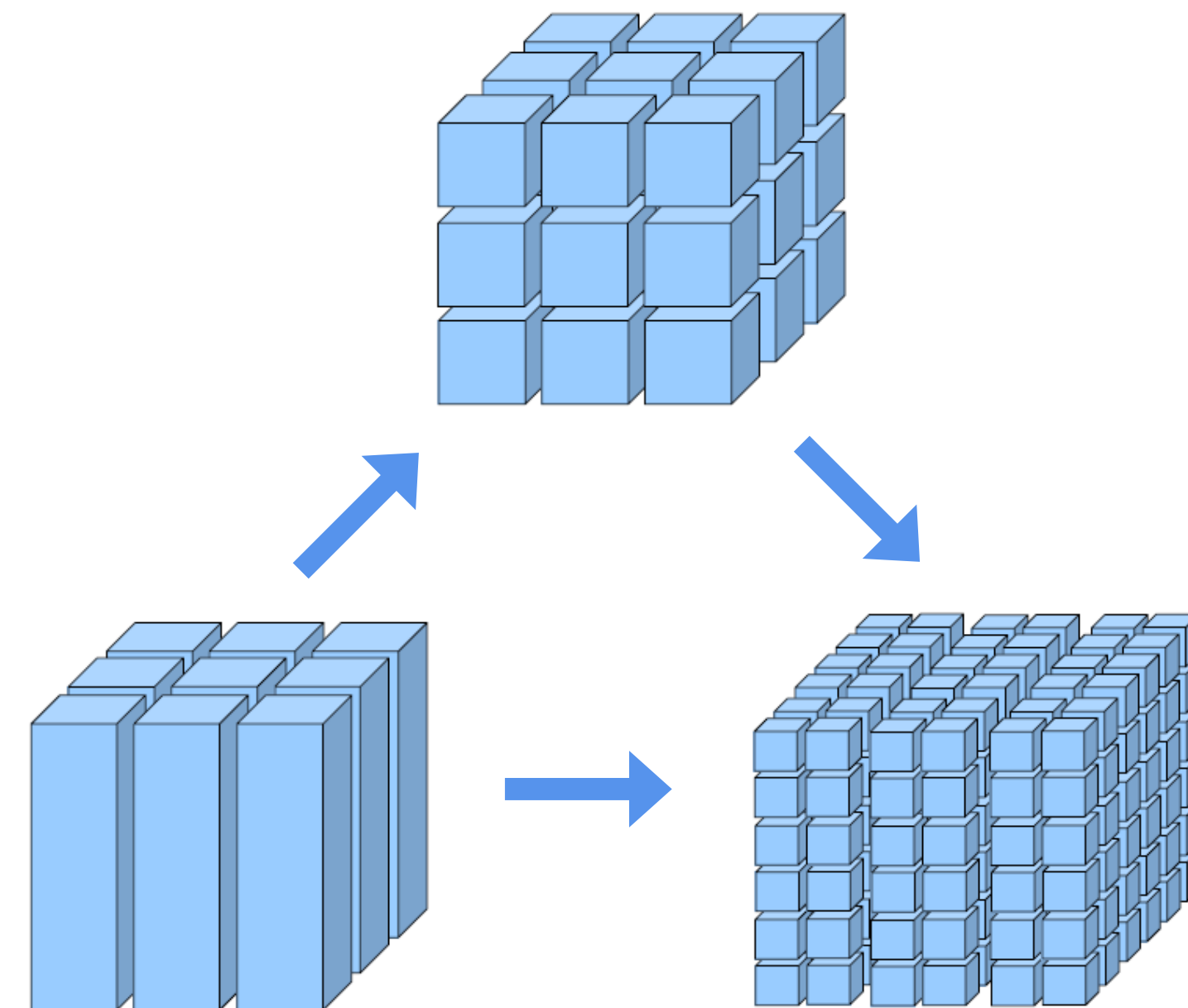
Roll-up and Drill-down

Roll-Up: Aggregates on hierarchies (less detailed cube)

Drill-Down: Disaggregates on hierarchies (more detailed cube)



Roll-Up Operations



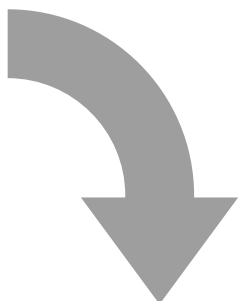
Drill-Down Operations

Roll-up and Drill-down

Roll-Up: Aggregates on hierarchies (less detailed cube)

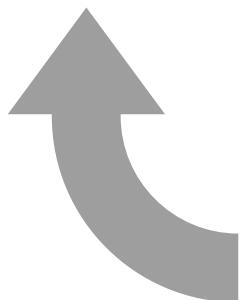
Drill-Down: Disaggregates on hierarchies (more detailed cube)

Time	North	Center	South
Jan 23	96,000	83,000	9,500
Mar 23	58,000	81,300	53,000
Jul 22	31,100	27,000	76,000
Feb 21	23,400	31,700	43,000
Feb 22	64,000	57,300	61,800
...	...	...	...



Roll-Up on Time Hierarchy
Drill-Down on Location Hierarchy

Time	FVG	Veneto	...	Lazio	...	Apulia	Sicily
Q1 23	6,000	3,500	...	3,500	...	3,100	6,200
Q1 22	1,200	4,500	...	7,300	...	4,500	3,800
Q2 22	1,700	4,200	...	3,200	...	3,700	3,100
Q3 22	5,200	1,600	...	8,200	...	4,100	2,600
Q4 22	6,400	5,700	...	2,300	...	7,800	5,300
...	...	...	...	...	...	...	...



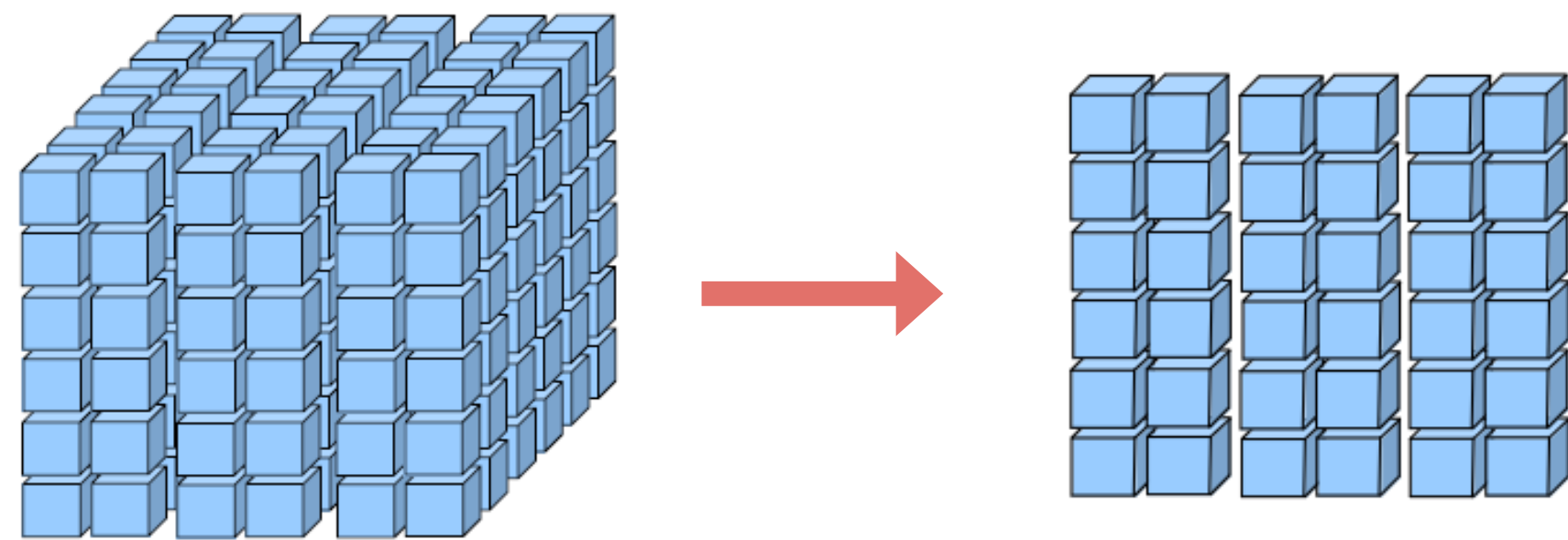
Roll-Up on Location Hierarchy
Drill-Down on Time Hierarchy

Slice and Dice

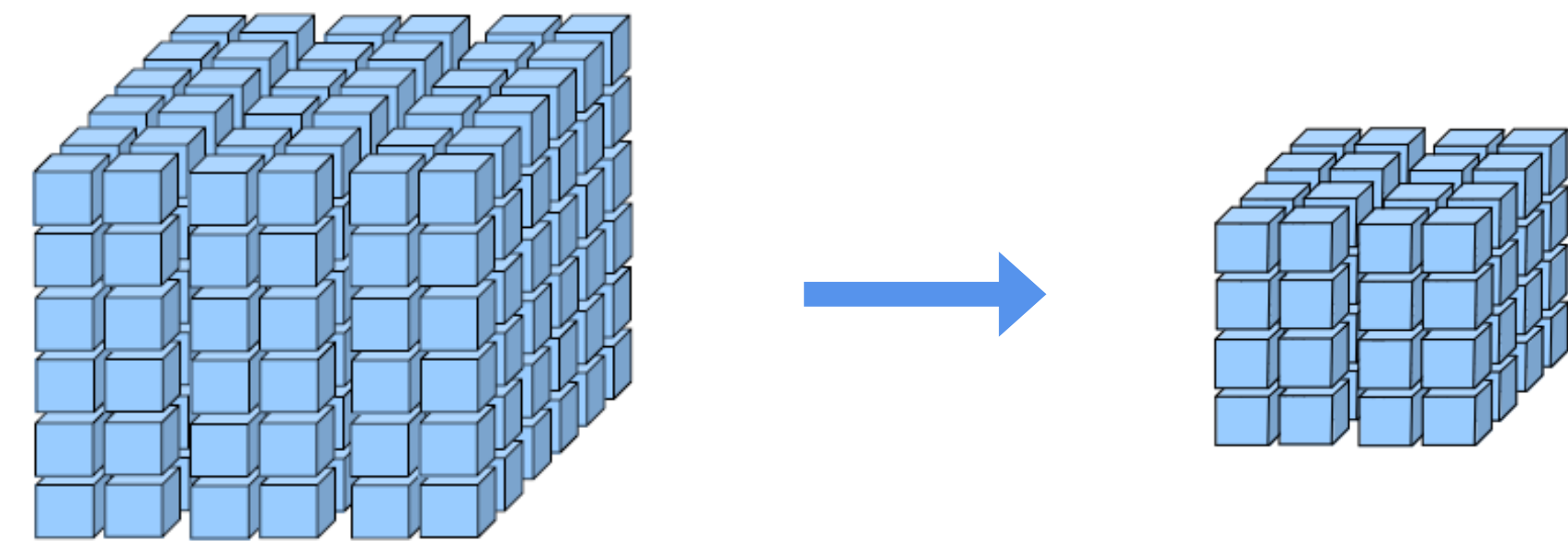
Slice: Reduces the number of dimensions by fixing one of them to a specific value

Dice: Uses selection criteria over a set of dimensions

Both slice and dice produce sub-cubes



Slice Operation



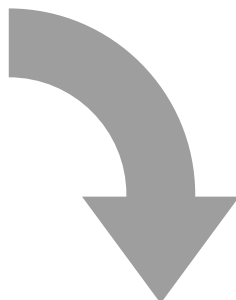
Dice Operation

Slice and Dice

Slice: Reduces the number of dimensions by fixing one of them to a specific value

Dice: Uses selection criteria over a set of dimensions

Quarter	Category	North	Center	South
Q1	Vegetables	96,000	83,000	9,500
	Meat	58,000	81,300	53,000
Q2	Vegetables	31,100	27,000	76,000
	Meat	23,400	31,700	43,000
Q3	Vegetables	64,000	57,300	61,800
...	...	...	...	...



Slice on Category='Vegetables'

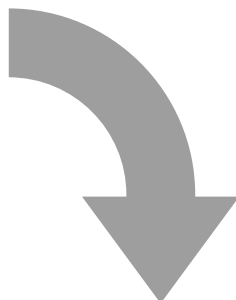
Quarter	Category	North	Center	South
Q1	Vegetables	96,000	83,000	9,500
Q2	Vegetables	31,100	27,000	76,000
Q3	Vegetables	64,000	57,300	61,800
...	...	...	...	...

Slice and Dice

Slice: Reduces the number of dimensions by fixing one of them to a specific value

Dice: Uses selection criteria over a set of dimensions

Quarte	Cat	FVG	Venet	...	Lazio	...	Sicily
Q1	Veg	6,000	3,500	...	3,500	...	3,100
	Meat	1,200	4,500	...	7,300	...	4,500
Q2	Veg	1,700	4,200	...	3,200	...	3,700
	Meat	5,200	1,600	...	8,200	...	4,100
Q3	Veg	6,400	5,700	...	2,300	...	7,800
...	...	...	...	...	...	...	...



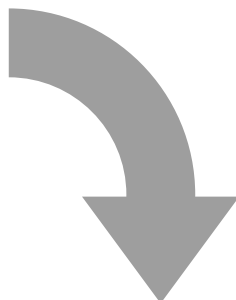
Dice on Category='Meat' $\wedge$ Area='Center'
 $\wedge$ Rev>1000

Quarter	Lazio	Umbria	Marche	...
Q1	7,300	4,100	2,500	...
Q2	8,200	3,700	1,800	...
Q3	1,300	2,400	2,100	...
...	...	...	...	...

Pivoting

Pivoting is a mere visual operation that does not change anything but the structure of the visualisation

Quarter	Category	North	Center	South
Q1	Vegetables	96,000	83,000	9,500
	Meat	58,000	81,300	53,000
Q2	Vegetables	31,100	27,000	76,000
	Meat	23,400	31,700	43,000
Q3	Vegetables	64,000	57,300	61,800
...	...	...	...	...



Pivoting on Category

Quarter	North		Center	
	Vegs	Meat	Vegs	Meat
Q1	96,000	58,000	83,000	9,500
Q2	31,100	23,400	27,000	31,700
Q3	64,000	45,000	57,300	51,000
...	...	...	...	...

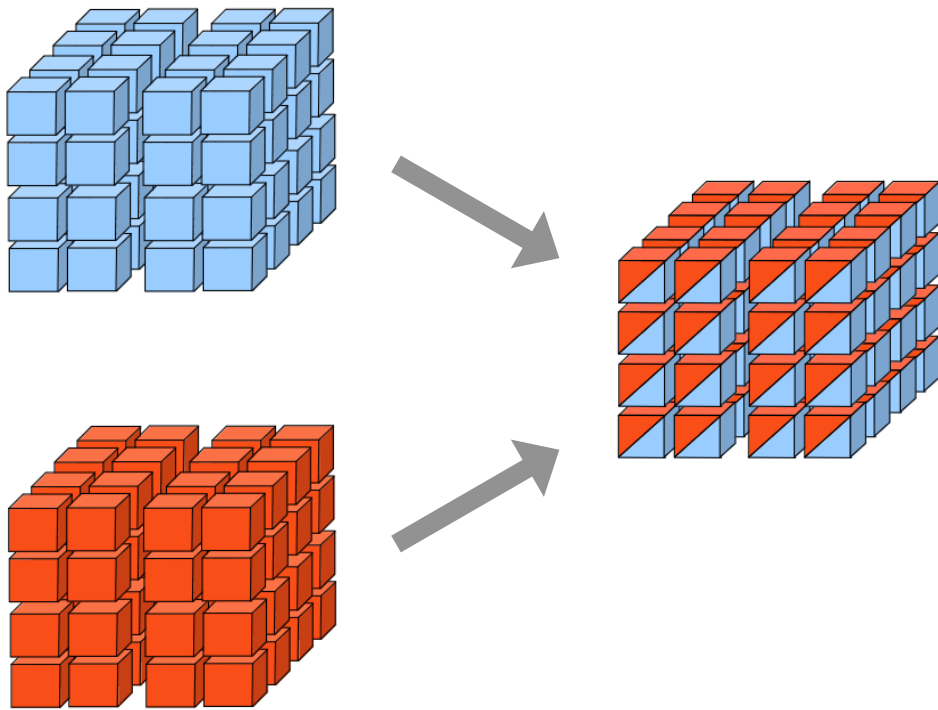
Drill-across

It is the operation of 'merging' different cubes with the same aggregation level

Time	North	Center	South
Jan 21	96,000	83,000	9,500
Mar 23	58,000	81,300	53,000
Jul 22	31,100	27,000	76,000
Feb 21	23,400	31,700	43,000
Feb 22	64,000	57,300	61,800
...	...	...	...

Time	North	Center	South
Jan 21	0.2	0.3	0.13
Mar 23	0.21	0.25	0.4
Jul 22	0.15	0.1	0.18
Feb 21	0.3	0.2	0.5
Feb 22	0.22	0	0.35
...	...	...	...

Time	North		Center		South	
	Disc	Rev	Disc	Rev	Disc	Rev
Jan 21	0.2	83,000	0.3	83,000	0.13	9,500
Mar 23	0.21	81,300	0.25	81,300	0.4	53,000
Jul 22	0.15	27,000	0.1	27,000	0.18	76,000
...	...	...	...	...	...	...



Data Warehousing in practice - Tools

Several tools exist to apply DW techniques

In some cases such tools address specific layers of the DW stacks (ETL, visualisations, BI tools, etc)

Non-exhaustive list of Data Warehousing tools:

- *Amazon Redshift*
- *Google Big Query*
- *Microsoft Power BI*
- *Pentaho*
- *Snowflake*
- *Tableau Public*
- *QlikView*



Tableau Public

Free (and limited) Data Warehousing platform

Focus on data visualisations for business intelligence

Allows to easily manipulate and understand the main components of the DFM model

Includes interactive dashboards and reports



[Why Tableau](#) ▾ [Products](#) ▾ [Solutions](#) ▾ [Resources](#) ▾ [Partners](#) ▾

[TRY NOW](#)

You'll be exploring in minutes

Create interactive graphs, stunning maps, and live dashboards in minutes. Save your viz to your Tableau Public profile, and share it anywhere on the web. Anyone can do it, it's that easy—and it's free.

[DOWNLOAD TABLEAU PUBLIC](#)

2023.1 AVAILABLE FOR WINDOWS AND MAC | [SYSTEM REQUIREMENTS](#)

Download Link:

tableau.com/products/public/download

(Sign Up required)

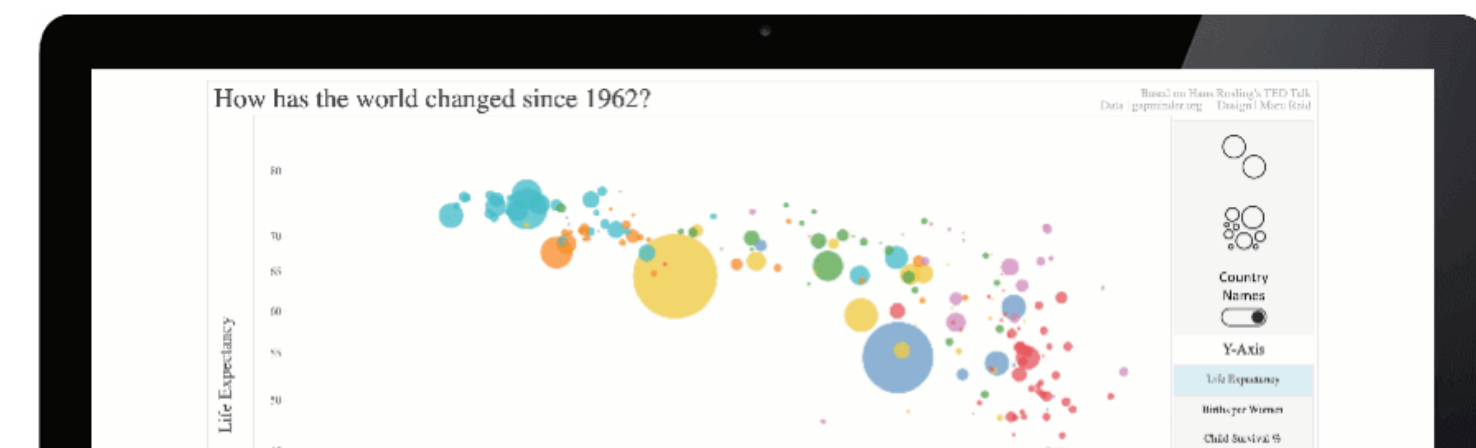


Tableau Workspace

Dimensions

Tables

Data

Abc Carnegie Classification 20...

Abc Control of institution

Abc County name

Abc Degree of urbanization (Ur...

Abc FIPS state code

Abc Geographic region

Abc Highest degree offered

Abc Historically Black College ...

ID number

Abc Level of institution

Abc Name

Abc Offers Associate's degree

Abc Offers Bachelor's degree

Abc Offers Doctor's degree - ot...

Abc Offers Doctor's degree - pr...

Abc Offers Doctor's degree - re...

Abc Offers Less than one year ...

Abc Offers Master's degree

Abc Offers One but less than t...

Abc Offers Other degree

Abc Offers Post-master's certi...

Abc Offers Postbaccalaureate ...

Abc Offers Two but less than 4 ...

Abc Religious affiliation

Abc Sector of institution

Abc State abbreviation

Abc Tribal college

Year

Abc ZIP code

ACT Composite 25th perc...

ACT Composite 75th perc...

Admissions total

Measures

Measure Values

SUM(Applicants total)

SUM(Bachelor's degr...

SUM(Master's degr...

SUM(Doctor's degr...

SUM(Part-time enroll...

SUM(Number of first...

BI Visualisations

For horizontal bars try

0 or more Dimensions

1 or more Measures

Cube (in tabular form)

Geographic region	Sector of institution	Applicants total	Bachelor's degrees aw..	Master's degrees aw..	Doctor's degrese - r..	Part-time enrollment	Number of first-time u..
Far West AK CA HI NV OR WA	Private not-for-profit, 4-y..	370,836	46,560	36,046	2,876	73,647	1,589
	Public, 4-year or above	1,047,737	183,295	42,147	6,217	175,172	2,584
Great Lakes IL IN MI OH WI	Private not-for-profit, 4-y..	541,518	81,229	41,203	2,164	139,682	1,739
	Public, 4-year or above	707,526	180,256	58,760	7,372	263,013	6,522
Mid East DE DC MD NJ NY PA	Private not-for-profit, 4-y..	1,163,397	131,729	95,393	6,204	248,300	5,558
	Public, 4-year or above	876,695	164,776	48,681	4,540	257,772	2,469
New England CT ME MA NH RI VT	Private not-for-profit, 4-y..	657,402	59,267	40,517	3,263	82,493	2,101
	Public, 4-year or above	236,176	46,829	12,120	1,027	74,421	699
Plains IA KS MN MO NE ND SD	Private not-for-profit, 4-y..	213,742	48,223	22,935	693	100,900	856
	Public, 4-year or above	263,387	87,068	24,054	3,531	153,167	2,353
Rocky Mountains CO ID MT UT WY	Private not-for-profit, 4-y..	54,582	21,137	8,839	228	30,875	182
	Public, 4-year or above	159,457	51,212	13,876	1,686	133,740	1,236
Southeast AL AR FL GA KY LA MS NC SC TN VA WV	Private not-for-profit, 4-y..	751,845	89,295	41,490	2,888	150,639	1,523
	Public, 4-year or above	1,137,213	291,417	91,204	10,136	452,011	5,891
Southwest AZ NM OK TX	Private not-for-profit, 4-y..	178,949	24,340	10,959	535	34,100	970
	Public, 4-year or above	440,211	133,110	47,093	5,134	232,327	2,153
US Service schools	Public, 4-year or above		201	4	0	27	

Workspace

DEMO

What are we going to do?

- Have a first look at Tableau Public
- See how to connect to data sources (and where to find them)
- Have a look at the Tableau Workspace components
- Understand how to create and manage hierarchies
- Navigate OLAP sessions through the various operations to make different analyses